

Screw Separator Cooling Jacket

My role is to simply remove a certain amount of energy from the surface of the wall. Everything is simulated as if the cooling jacket is a concurrent pipes.

The primary change is that the path length of the helical coil will have to increase as time goes on to simulate the increasing diameter of the screw, and this will have to be reflected by the pitch of this change.

```
> restart;

> print("====Key Equations====");
eq1 := Q__req = m*Cp*(T__cold2 - T__cold1); #Energy Balance from
page 478 fundametals of heat and mass transfer (Shows energy
extraction)
eq2 := Q__req = h__j*A*DeltaT__lm;          #Page 483 fundametals
of heat and mass transfer
DeltaT__lm := ((T__hot1 - T__cold2)-(T__hot2 - T__cold1))/(ln(
(T__hot1-T__cold2)/(T__hot2-T__cold1))); #Page 482 fundametals of
heat and mass transfer
```

"====Key Equations===="

$$eq1 := Q_{req} = m C_p (T_{cold2} - T_{cold1})$$

$$eq2 := Q_{req} = h_j A \Delta T_{lm}$$

$$\Delta T_{lm} := \frac{T_{hot1} - T_{cold2} - T_{hot2} + T_{cold1}}{\ln \left(\frac{T_{hot1} - T_{cold2}}{T_{hot2} - T_{cold1}} \right)} \quad (1.1)$$

```
> #1 is inlet
#2 is outlet
#Hot is wall
#Cold is the fluid

print("====Specifications====");
T__cold1 := 281.001822 + 273.15; #Kelvin #depends on temperature
in #NEED SPECIFYING
m := 0.25; #kg/s
```

```

T_hot1 := 500 + 273.15; #Kelvin #depends on energy into
separator #NEED SPECIFYING
T_hot2 := 285 + 273.15; #Kelvine # The desired temperature of
wall at end of separator. #Need specifying

```

```

#Constant diameter separator
L_jacket := 1.2; #m #Length of Screw Conveyor
D_separator := 0.1016; #m
GAP := 0.01; #m (10 mm annular gap)
D_jacket := D_separator + 2*GAP;

```

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Q_req := 1100; #W #Required energy to be extracted

```

```

print ("====Properties of Helisol 5A, tube side====");
rho := 689.5; #kg/m3
Cp := 2150; #J/kg/K
mu := 0.41e-3; #Pa/s
k_fluid := 0.076 #W/m/K

```

"====Specifications===="

$T_{cold1} := 554.151822$

$m := 0.25$

$T_{hot1} := 773.15$

$T_{hot2} := 558.15$

$L_{jacket} := 1.2$

$D_{separator} := 0.1016$

$GAP := 0.01$

$D_{jacket} := 0.1216$

$Q_{req} := 1100$

"====Properties of Helisol 5A, tube side===="

$$\rho := 689.5$$

$$C_p := 2150$$

$$\mu := 0.00041$$

$$k_{fluid} := 0.076$$

(1.2)

> #helical cannels (If pressure is too high, consider splitting channels into two, but will half velocity and Re.)

w := 0.03; #m #Channel width

t := 0.02; #m #Wall thickness

P := w + t; #m #pitch

N__turn := L__jacket/P; #number of turns

L__turn := ((Pi*D__separator)**2 + P**2)**(1/2); #m #Length per turn

L__coil := N__turn * L__turn; #m

A__c := GAP*w;

D__h := 2*GAP*w/(GAP +w);

D__coil := (D__jacket + D__separator)/2;

$$w := 0.03$$

$$t := 0.02$$

$$P := 0.05$$

$$N_{turn} := 24.00000000$$

$$L_{turn} := 0.3230782933$$

$$L_{coil} := 7.753879039$$

$$A_c := 0.0003$$

$$D_h := 0.01500000000$$

$$D_{coil} := 0.1116000000$$

(1.3)

> T__cold2 := solve(eq1,T__cold2);

v := m/(rho*A__c); #m/s

A := Pi * D__separator*L__jacket;

$$T_{cold2} := 556.1983336$$

$$v := 1.208605270$$

$$A := 0.3830229764$$

(1.4)

> Re__1 := rho*v*D__h/mu;

(1.5)

$$\Re_j := 30487.80490$$

(1.5)

```
> Pr := (Cp *mu)/k__fluid;
#cooling, Page 491, fundametals of heat and mass transfer
#Updated to use Seban and McLaughlin correlation(For helical
coils, not channeled jackets, is this okay?)
Nu := 0.023*Re__1**0.85*Pr**0.3*(D__h/D__coil)**0.1;
h__j := Nu*k__fluid/D__h; #heat transfer coefficient (Book uses
U)
```

$$Pr := 11.59868421$$

$$N := 254.3316407$$

$$h_j := 1288.613646$$

(1.6)

```
> print("====Summary====");
DeltaT__lm := DeltaT__lm;
Re__1 := Re__1;
v := v; #m/s
T__cold1 := T__cold1; #kelvin
T__cold2 := T__cold2; #kelvin
T__hot1 := T__hot1; #Kelvin
T__hot2 := T__hot2; #Kelvin
print("=====");
```

"====Summary===="

$$\Delta T_{lm} := 53.32054124$$

$$\Re_j := 30487.80490$$

$$v := 1.208605270$$

$$T_{cold1} := 554.151822$$

$$T_{cold2} := 556.1983336$$

$$T_{hot1} := 773.15$$

$$T_{hot2} := 558.15$$

"=====

(1.7)

```
> # Required Area vs Available Area
A__req := Q__req / (h__j * DeltaT__lm);
A__av := 3.14 * D__separator * L__jacket;

# Check if design is valid
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```

if A__av > A__req then
    print("Design Successful");
else
    print("Design Failed");
end if;

```

$$A_{req} := 0.01600941306$$

$$A_{av} := 0.3828288$$

"Design Successful"

(1.8)

```

> #Pressure calculations(Used the Mishra & Gupta correlation)

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```

> f__c := 0.3164/Re__1^0.25 + 0.03*(D__h/D__coil)^0.5;

```

$$f_c := 0.03494298748$$

(1.9)

```

> #Pressure added due to pumping up system (Against gravity)

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```

DP__friction := 4*f__c*(L__coil/D__h)*((rho*v**2)/2); #Pa

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DP__total := DP__friction; #Pa

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DP__kPa := DP__total/1000; #kPa

```

$$DP_{friction} := 36384.88686$$

$$DP_{total} := 36384.88686$$

$$DP_{kPa} := 36.38488686$$

(1.10)